Post-Mid-Semester Research Assignment

Prompts 03 and 04 — Research Progress Record

Research topic: Evaluating Tool-Selection Accuracy in Multi-Tool Agentic AI Systems

Candidate study: Tool-Selection Robustness Under Dynamic and Confusable Tool Inventories

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Course: AI Tools for Research

Coverage: Prompt 03 (literature-to-experiment mapping) and Prompt 04 (feasibility and novelty verification).

Record boundary: Consolidation of the responses already produced. No new source verification, code execution, measurements, or later prompts are represented here.

Evidence convention

VERIFIED = documented earlier primary-source/repository check or direct available record, with scope specified. INFERRED = research-planning judgment. UNKNOWN = not established / pending verification. Public repository visibility does not prove runnable Colab access.

1. Prompt 03 — Literature-to-Experiment Mapping

Exact submitted Prompt 03

My literature review Evaluating Tool-Selection Accuracy in Multi-Tool Agentic AI Systems: A Survey of Metrics, Benchmarks, and Open Challenges identified these candidate gaps:
Gap 1 — Tool selection under dynamic tool-set changes
Gap 2 — Candidate-set composition and similar-tool confusion
Gap 3 — Isolating selection errors from downstream failures
Gap 4 — Functional tool retrieval versus textual similarity
Gap 5 — Agent-as-tool selection and delegation reliability

Identify three published methods from my review that could realistically be implemented or reproduced. For each method, map its original paper to the algorithm, dataset, code or model access, evaluation metric, known limitations and computational requirement. Distinguish verified access to code/weights from unverified suggestions. Compare three possible experimental studies: (A) reproduction plus extension, (B) controlled comparison, and (C) targeted modification. For each study state the smallest working baseline, the testable question, expected resource burden and evidence required. Recommend one minimum viable research experiment for Google Colab. Provide a literature-to-experiment mapping table and list what I must personally verify before coding. Do not propose an elaborate new architecture without a scientific rationale.

3.1 Original paper → method → executable candidate

Study	Method / algorithm	Data / metrics	Availability and verification	Colab burden
MetaTool / ToolE — Huang et al., ICLR 2024	Tool necessity and discriminative selection among candidate tools; prompt construction and label matching.	ToolE; Correct Selection Rate / selection correctness; examples include confusable, no-tool and multi-tool settings.	VERIFIED in earlier response: public repository, all_clean_data.csv, multi_tool_query_golden.json, plugin_des.json, prompt-building scripts. Original model weights as one package: UNKNOWN.	High feasibility for subset and lightweight selectors; full original model replication lower.
ToolRet — Shi et al., Findings ACL 2025	Sparse and dense retrieval/ranking against natural-language tool descriptions; trained retrievers as heavier option.	About 7,600 retrieval tasks and 43,000 tools as reported; Recall@k, nDCG and ranked-retrieval measures.	VERIFIED in earlier response: official code/data-resource documentation. Exact checkpoint, license and downloads: pending.	BM25 CPU practical; small embedding model practical in principle; full training heavier.
TaskBench / TaskEval — Shen et al., NeurIPS 2024	Task automation as graph prediction: tool nodes and dependency edges.	Graph tasks; Node-F1, Edge-F1 and normalized edit distance.	VERIFIED publication links Microsoft JARVIS; clean Colab installation, data packaging and compatibility UNKNOWN.	Graph scoring inexpensive; generating complete trajectories more demanding.

Scope qualification: A retriever choosing a top-ranked tool is not identical to a native LLM tool-calling agent. A subset evaluation is not an exact reproduction of published model scores.

3.2 Experimental pathways A, B and C

Pathway	Smallest working baseline	Question / tests	Resource and evidence need
A — Reproduction + extension	MetaTool task/evaluation with one selector.	Does selection change when tools are replaced or deprecated? Original versus changed inventory.	Moderate; original prompts/settings/model may be needed for exact replication. Need documented reproduction and paired extension.
B — Controlled comparison (recommended)	BM25 and dense retrieval on the same labeled cases.	Which method retains accuracy under tool-inventory updates and confusable alternatives?	High feasibility; same queries, inventory sizes, ground truth, uncertainty and paired logs.
C — Targeted modification	BM25 plus one capability-aware reranking rule.	Does the rule improve selection on ambiguous descriptions?	Moderate; isolated component change, fixed baseline, ablation and leakage control.

3.3 Selected minimum viable study — planning only

Recommendation (INFERRED): Pathway B; MetaTool/ToolE as the primary evaluation source and BM25 versus a lightweight dense retriever as the initial comparison. Gap 1 (dynamic tool sets) remains primary, with Gap 2 (confusability) as a controlled factor. Gap 4 informs the retrieval comparator; TaskBench is a secondary diagnostic reference.

Prompt 03 research question: How do changes in candidate-tool availability and functional overlap affect lightweight tool-selection methods, and can retrieval over updated descriptions reduce errors?

Condition	Planned change	Validity constraint
C0 original	Unmodified tool inventory and verified correct tool.	Use authentic, documented labels.
C1 added	Add plausible but irrelevant distractor tools.	Keep question and scoring rule fixed.
C2 similar	Add overlapping or similar candidate tools.	Allow multiple valid choices when warranted.
C3 changed registry	Remove/deprecate a tool and replace with similar functionality.	Relabel accepted answers independently; replacement equivalence cannot be assumed.

Initial specification: Approximately 100–200 single-tool queries (planning estimate); candidate sizes 5/10/20 only where legitimate; primary top-1 accuracy; secondary paired degradation, invalid/deprecated-tool rate, Recall@k where applicable, subgroup errors, latency and confidence intervals. CPU first, optional GPU for embedding model. None of these figures is an observed result.

3.4 Prompt 03 outstanding verification

Before coding: confirm MetaTool and ToolRet evaluation rules, split/labels, exact file formats, reuse rights, checkpoint identity/license/access, Colab memory and dependencies, paired replacement ground truths, seed/candidate ordering and leakage controls. Reproducing published system numbers is not yet established.

2. Prompt 04 — Feasibility and Novelty Assessment

Exact submitted Prompt 04

My candidate study is Tool-Selection Robustness Under Dynamic and Confusable Tool Inventories
Pathway B · MetaTool as primary evaluation source · BM25 and a small dense retriever as comparison methods. Develop a feasibility and novelty-verification plan. Provide scholarly search strings and specific categories of neighboring work to inspect, without inventing titles, DOIs or URLs. Create a matrix with columns: published method, VERIFIED contribution (with source), limitation, proposed difference and evidence still missing. Check whether the dataset, code, model weights, license and benchmark protocol are accessible; flag all unresolved availability claims. Estimate computational and time constraints for Google Colab. Narrow the study to a viable experimental question and recommend a fallback if the preferred repository, weights or dataset are unavailable.

4.1 Feasibility status and neighboring research

Conditional GO (INFERRED): A lightweight Colab comparison appears feasible, but baseline reproducibility, dataset/license suitability, checkpoint access and novelty are not established. Previously surfaced neighboring work: BiasBusters (2025) studies equivalent-tool bias/metadata and perturbations; Too Many Tools, Too Much Confusion? (2026) addresses scalability and semantic collisions; AutoTool (ICML 2026) addresses evolving tool selection; ToolRet addresses retrieval comparison. These findings were reported in the Prompt 04 response; no additional verification has been performed for this PDF.

4.2 Novelty/feasibility matrix

Published method	Earlier VERIFIED contribution (source)	Limitation classification	Proposed difference	Still missing
MetaTool / ToolE	Tool awareness and confusable selection benchmark (official ICLR paper/repository).	Review-reported boundary: lacks controlled add/retire studies in the proposed format.	Paired original/changed inventories.	Split, labels, transformed-data rights and overlap with prior perturbation work.
ToolRet	Benchmarks tool retrieval and provides training/evaluation resources (ACL 2025 / repository).	Original reported finding: ordinary IR transfer can be weak for tools.	Controlled version change and similarity conditions for lightweight methods.	Exact evaluation settings, checkpoint and representativeness.
Gorilla	Retriever-aware API-call generation for documentation changes (Patil et al.).	Reviewer inference: not every registry-churn condition is covered.	Isolate retrieval correctness under replacement.	Compare to original adaptation protocol.
BiasBusters	Equivalent-tool bias, metadata, description and position effects (2025 paper).	Inferred boundary: choice bias versus paired registry churn.	Interaction of tool replacement and confusability.	Full protocol and redundancy assessment.
AutoTool	Training-based selection in evolving tool settings (2026 publication).	Inferred boundary: differs from untrained retriever comparison.	Low-resource untrained evaluation under registry update.	Check its ablations for identical controls.

4.3 Scholarly search strategy already proposed

- "tool selection" AND ("dynamic toolset" OR "evolving tool inventory" OR "tool registry")
- ("tool retrieval" OR "API retrieval") AND ("deprecated" OR "updated" OR "replacement")
- "tool selection" AND ("similar tools" OR "semantic collision" OR "confusable tools")
- ("tool calling" OR "function calling") AND ("description perturbation" OR "metadata robustness")
- ("BM25" AND "dense retrieval") AND ("tool retrieval" OR "API selection")

• "tool selection" AND ("candidate set size" OR "distractor composition")

• ("tool retrieval" OR "tool routing") AND ("temporal robustness" OR "version drift")

• ("dynamic tool discovery" OR "MCP tool discovery") AND evaluation

Categories: dynamic API adaptation; retrieval/routing; equivalent/confusable tools; robustness perturbations; temporal evaluations; retrieval-to-execution distinction; MCP/runtime registries. Suggested discovery venues: Google Scholar, Semantic Scholar, ACL Anthology, arXiv, and ICLR/ICML/NeurIPS proceedings. This is an identified search strategy, not proof that a comprehensive search was completed.

4.4 Data, code, model and licensing audit

Resource	Earlier status	Pending check
MetaTool code / listed data	VERIFIED public repository and named dataset/description files.	Parsing and runtime compatibility.
MetaTool repository license	VERIFIED at repository level: MIT, per earlier response.	Separate dataset origins and redistribution.
ToolRet evaluation code / data listings	VERIFIED public documentation, per earlier response.	Local download, dataset conditions and exact benchmark protocol.
ToolRet repository license	VERIFIED at repository level: Apache 2.0, per earlier response.	Licenses of individual datasets and models.
BM25	INFERRED feasible with Python library.	Install/run in Colab; version freeze.
Small dense model	Earlier example identifier: intfloat/e5-small-v2.	Checkpoint/revision/license/weights access and runtime UNKNOWN.
Google Colab GPU	Not guaranteed; CPU fallback anticipated.	Actual memory, storage, acceleration and timing UNKNOWN.
Published score reproduction	UNKNOWN — no baseline run.	Document reproducible configurations and outputs.

4.5 Narrowed experimental question and controls

Recommended RQ (INFERRED): How does replacing an available tool with a functionally similar alternative affect top-1 retrieval-based selection accuracy, and does this effect differ between BM25 and a pretrained dense retriever?

Compare an original inventory with a matched replacement condition for the same single-tool requests, fixed candidate count, and independently verified version-specific accepted labels. Secondary metrics: Recall@3, invalid selection rate and latency; paired confidence intervals and matched error analysis proposed. An unchanged control tool is recommended. Actual replacement-pair availability is UNKNOWN.

4.6 Google Colab effort estimate — inferred, not observed

Activity	Estimate	Resource
Dataset inspection	1–2 hours	CPU
Label inspection / eligible cases	2–4 hours	CPU
BM25 setup/validation	1–3 hours	CPU
Dense checkpoint setup/evaluation	2–4 hours	CPU or optional GPU
Replacement-case validation	4–8 hours	Manual + CPU
Controlled evaluations	1–3 hours	CPU at small scale
Analysis	2–4 hours	CPU
Reproducibility documentation	2–4 hours	Minimal compute

Total planning estimate: approximately 15–32 hours (3–6 working days), excluding major access problems; not a measured runtime or schedule commitment. Pilot suggested at roughly 100–200 eligible requests with small controlled candidate pools.

4.7 Risks and fallback

Blocker	Existing fallback	Consequence
MetaTool scripts fail	Read available structured files directly and score transparently.	Subset reuse, not exact original system reproduction.
ToolE unavailable	Use license-verified ToolRet subset.	Dataset and protocol revision required.
Valid replacement pairs scarce	Study addition of similar distractors instead.	Emphasis shifts to candidate-set composition.
Chosen model weights unavailable	Select another small accessible checkpoint with documented license.	Comparator changes; record it.
Dense model does not run	Use TF-IDF cosine as second baseline.	No longer sparse versus neural dense.
Redistribution rights unclear	Release identifiers/scripts/transformation rules, not restricted data.	Reproduction requires authorized original data access.

Go/no-go: GO for source and resource feasibility checks; NO-GO for full study until labels/replacements, legal access, both baseline predictions, and defensible literature differentiation are demonstrated. No new model, benchmark or novel contribution is claimed.

3. Research Progress Summary

Completed in Prompts 03–04: Mapped three published methods; compared three contribution pathways; selected Pathway B and a lightweight Colab configuration; identified a research question, metrics and matched conditions; recorded neighboring-paper concerns, accessibility statuses, risk controls, fallbacks, rough resource requirements and conditional approval gates.

Not completed: dataset download/inspection in Colab, runnable BM25 or dense model, verified replacement labels, power/sample-size justification, independent license evaluation, full systematic novelty determination, frozen experiment protocol, execution logs, statistical outcomes or implementation-paper results.

Recommendation continuity: Prompt 03 proposed broader inventory additions, similarity, and replacements; Prompt 04 subsequently narrowed emphasis to functionally equivalent replacement. This is a documented refinement, not evidence that the experiment has been implemented. Dynamic inventory changes remain the central direction.

4. Research Decision and AI-Assistance Record

Record	What was documented	Status / next action
D008 — Prompt 03 recommendation	Pathway B; MetaTool-based controlled BM25 vs dense comparison under dynamic and confusable inventories.	INFERRED; inspect data, permissions and establish baseline.
D009 — Prompt 04 refinement	Narrow to matched replacement robustness; candidate-size control and condition-specific labels.	INFERRED; validate replacement pairs and relative novelty.
Computing constraint	Google Colab is the user-specified available environment.	User-confirmed; actual tier resources UNKNOWN.
AI assistance	ChatGPT structured literature-to-experiment mapping, feasibility matrix, search strings, estimates, and go/no-go advice.	Planning only; no tests, experimental outputs or independent human verification claimed.
Prior decision-log file	Existing research_decisions.md contains D001–D007; D008/D009 were proposed in conversation.	Do not retroactively relabel them accepted or recorded to file.

Provenance note: The accessible research_decisions.md contained D001–D007, but not D008 or D009. The latter are quoted here as proposals from the Prompt 03/04 responses; the log has not been silently rewritten.

5. References and resource links used in Prompts 03–04

1. Huang et al. MetaTool Benchmark for Large Language Models: Deciding Whether to Use Tools and Which to Use. ICLR 2024. https://proceedings.iclr.cc/paper_files/paper/2024/hash/bc12914d66b41b6bfc2d3a5decdb498b-Abstract-Conference.html ; https://github.com/HowieHwong/MetaTool

2. Shi et al. Retrieval Models Aren’t Tool-Savvy: Benchmarking Tool Retrieval for Large Language Models. Findings of ACL 2025. https://aclanthology.org/2025.findings-acl.1258/ ; https://github.com/mangopy/tool-retrieval-benchmark

3. Shen et al. TaskBench: Benchmarking Large Language Models for Task Automation. NeurIPS 2024. https://proceedings.neurips.cc/paper_files/paper/2024/hash/085185ea97db31ae6dcac7497616fd3e-Abstract-Datasets_and_Benchmarks_Track.html ; https://github.com/microsoft/JARVIS

4. Patil et al. Gorilla: Large Language Model Connected with Massive APIs (2023). Earlier review reference; DOI/arXiv identifier as recorded in review: https://arxiv.org/abs/2305.15334

5. BiasBusters: Uncovering and Mitigating Tool Selection Bias in Large Language Models (2025). https://arxiv.org/abs/2510.00307

6. Too Many Tools, Too Much Confusion? Navigating Agentic Tool Selection at Scale (2026). https://www.mdpi.com/1999-4893/19/6/447

7. AutoTool: Dynamic Tool Selection and Integration for Agentic Reasoning (2026). https://proceedings.mlr.press/v306/zou26p.html

8. Practitioner dynamic tool-discovery comparison (2026), described in the Prompt 04 response; full bibliographic details were not recorded and are NOT ESTABLISHED.

9. Model identifier cited as candidate: intfloat/e5-small-v2. Weights/revision/license access not verified.

Source boundary: Bibliographic details reflect those already present in Prompt 03/04 outputs and the earlier review citation named there; no independent reference expansion or verification occurred during PDF assembly.